INMO(33rd) – 2018

Time – 4 Hrs

Instructions:

- **Calculators (in any form) and protractors are not allowed.**
- **Ruler and compass allowed.**
- **Answer all the questions.**
- **All Questions carry equal marks. Maximum marks: 102.**
- **Answer to each question should start on a new page. Clearly indicate the question number.**

1. Let ABC be non-equilateral triangle with integer sides. Let D and E be respectively the mid-points of BC and CA; let G be the centroid of triangle ABC. Suppose D, C, E, G are concyclic. Find the least possible perimeter of triangle of ABC.
2. For any natural number n, consider a $1 \times n$ rectangular board made up of n unit squares. This is covered by three types of tiles: 1×1 red tile, 1×1 green tile and 1×2 blue domino. Let t_n denotes the number of ways covering $1 \times n$ rectangular board by these three types of tiles. Prove that t_n divides t_{2n+1}.
3. Let Γ_1 and Γ_2 be two circles with respective centres O_1 and O_2 intersecting in two distinct points A and B such that $\angle O_1AO_2$ is an obtuse angle. Let the circumcircle of triangle O_1AO_2 intersects Γ_1 and Γ_2 respectively in points C and D. Let the line CB intersects Γ_2 in E; let the DB intersects Γ_1 in F. Prove that the points C, D, E, F are concyclic.
4. Find all polynomials with real coefficients $P(x)$ such that $P(x^2+x+1)$ divides $P(x^3-1)$.
5. There are $n \geq 3$ girls in a class sitting around a circular table, each having some apples with her. Every time the teacher notices a girl having more apples than both of her neighbours combined, the teacher takes away one apple from that girl and gives one apple each to her neighbours. Prove that this process stops after a finite number of steps. (assume that the teacher has an abundant supply of apples).
6. Let N denote the set of all natural numbers and let $f: N \to N$ be a function such that:
   (a) $f(mn) = f(m)f(n)$ for all m, n in N.
   (b) $m+n$ divides $f(m)+f(n)$ for all m, n in N.

   Prove that there exists an odd number k such that $f(n) = n^k$ for all n in N.

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